The Shifting Peak: Analyzing Diurnal Maximum Temperature Trends in Épinal (Grand Est, France)

Rong Martin-Siebler ZHOU

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1 Abstract

Background: The diurnal temperature cycle is fundamental to ecology, agriculture, and human health. While changes in diurnal temperature *range* (DTR) are well documented, shifts in the *timing* (phase) of the daily maximum temperature remain poorly characterized at local scales, particularly across meteorological seasons.

Methods: Using 35 years (1992–2025) of hourly temperature records from the Météo-France RADOME station at Épinal, France (48.21 °N, 6.45 °E), I extracted the clock hour of daily maximum temperature from 12,163 valid days and modelled its long-term trend using a von Mises circular regression by direct maximum likelihood (Fisher & Lee 1992). Confidence intervals were obtained from 1,000 bootstrap resamples. A year–season interaction likelihood-ratio test formally tested whether trends differ across meteorological seasons; because it was significant ($\chi^2 = 13.71$, df $= 3$, $p = 0.0033$), seasonal stratification is the primary analysis framework. Per-season estimates from the interaction model are reported with Benjamini-Hochberg FDR correction. Pooled (all-season) results and six sensitivity variants are presented as supplementary context.

Results: Winter (DJF) shows a significant advance of −0.19 h/decade (95% bootstrap CI: −0.32, −0.05; Wald $p = 0.0015$, BH-rejected), corresponding to approximately −11 min/decade. The pooled estimate of −0.03 h/decade masks opposing winter and autumn (+0.09 h/decade) trends (autumn bootstrap CI: +0.007 to +0.170), illustrating a Simpson's paradox. Spring (−0.07 h/decade) and summer (+0.02 h/decade) are near zero and non-significant. The winter advance is directionally robust across all six sensitivity variants.

Conclusions: The timing of the daily maximum temperature at Épinal shows significant seasonally heterogeneous trends. Winter afternoons are measurably cooling earlier in the day over the past three decades, while the annual average hides this signal. These findings highlight the importance of seasonal stratification for detecting shifts in the diurnal temperature phase, with implications

for heat-health warning systems, crop modelling, and building energy demand forecasting.

2 Introduction

The diurnal temperature cycle – the rise from daily minimum to daily maximum and subsequent decline – is a fundamental climatic rhythm that shapes ecological processes, agricultural productivity, human health outcomes, and energy demand [Wang and Dillon, 2014]. Most research on changes in the diurnal cycle has focused on the diurnal temperature range (DTR), documenting a narrowing of the range in many regions as minimum temperatures have risen faster than maximum temperatures [Easterling et al., 1997]. Far less attention has been paid to whether the *phasing* (timing) of the diurnal cycle is changing – that is, whether the clock hour at which the daily maximum temperature occurs has shifted over time.

Recent evidence suggests that the phase of the diurnal temperature cycle may indeed be shifting at global scales. In a conference presentation, Shmuel et al. [2026] analysed historical hourly temperature records spanning the 1980s to the near present and documented systematic shifts in the timing of both morning warming and afternoon cooling periods. Their key findings include morning warming-period temperatures occurring approximately 15 minutes earlier per decade globally, and afternoon cooling-period temperatures occurring more than 20 minutes later per decade. Under a moderate emissions scenario (SSP2-4.5), they project that equivalent temperatures could shift by over three hours by 2100 relative to late-20th-century baselines.

In a complementary peer-reviewed global analysis, Wang and Dillon [2014] used over 1.4 billion hourly measurements from 7,906 weather stations worldwide covering the period 1975–2013. They reported that diurnal temperature variation increased by approximately 1.0,°C in temperate regions, with this increase particularly pronounced in Europe, including France. They further documented coherent changes in the phase of the *annual* temperature cycle, suggesting that both diurnal and seasonal timing shifts are occurring in response to radiative forcing.

Together, these findings suggest that the diurnal temperature cycle is undergoing systematic changes in both magnitude and phase at global and continental scales. This study aims to explore the validity of this global trend at the local level, namely, in the city of Épinal (48° 10’ 28” N, 6° 27’ 04” E), Grand Est region, France.

2.1 Present study

This study seeks to answer a simple question: *Has the timing of the diurnal temperature peak in city of Épinal changed over the past three and a half decades?* Using 35 years of hourly temperature records from a single well-instrumented Météo-France RADOME station, I apply a von Mises circular regression model – the appropriate statistical framework for a circular response variable (0–23 hours) – to estimate the trend in peak temperature timing. A year–season interaction likelihood-ratio test determines whether seasonal stratification is warranted; per-season estimates with false discovery rate control constitute the primary analysis, while the pooled (all-season) estimate and sensitivity variants are presented as supplementary context.

This study thus compares the local-scale evidence from a single station in eastern France with the global phase shifts provisionally identified by Shmuel et al. [2026], at the resolution of the daily temperature peak.

3 Methods

3.1 Data source

Sub-daily (hourly) temperature records were obtained from the Météo-France Synop Essentielles / RADOME network, station **88136001** (Épinal, 48.21°N, 6.45°E, 317 m a.s.l.). The station opened on 1 June 1986; no sub-daily data exist before this date. After quality-control filtering (see below), the analysis period spans 1 January 1992 to 31 December 2025, yielding 35 years of valid hourly records. Raw data were acquired via the Météo-France public data API using the `pooch` data registry and archived on OSF for reproducibility.

3.2 Study area

Épinal (48.21°N, 6.45°E) is situated in the Moselle River valley at the western foothills of the Vosges Mountains, at the transition between the Lorraine plain and the Vosges massif (elevation 317 m a.s.l.). The local climate is temperate oceanic (Köppen: *Cfb*), characterised by mild temperatures and precipitation distributed throughout the year [2022]. The Vosges orographic barrier produces a pronounced west–east precipitation gradient, with the western slopes receiving substantially more precipitation than the rain-shadowed Alsatian plain to the east via the föhn effect [2013]. Valley topography and forest cover may influence local boundary-layer development and, consequently, the timing of the diurnal temperature maximum.

3.3 Software

All processing was performed in Python 3.12. The data acquisition employed the `pooch` library. The data processing employed the `numpy`, `pandas`, `scipy`, and `statsmodels` libraries. The graphing was performed with `matplotlib` and `seaborn`. Reproducibility is ensured via `pyproject.toml`-pinned dependencies and the pipeline `Makefile`. Timezone was handled by native `zoneinfo`.

3.4 Preprocessing and quality control

UTC to local time conversion. Raw UTC timestamps were converted to local time (CET/CEST) via `zoneinfo.ZoneInfo("Europe/Paris")`, with correct handling of 23-hour (spring-forward) and 25-hour (autumn-back) daylight saving transition days via the `fold` attribute.

Missing data filter. A calendar day was retained only if at least 75% of hourly observations were non-null (i.e., ≤ 6 missing hours out of 24).

Diurnal amplitude filter. Days with a diurnal temperature range $T_{\max} - T_{\min} < 2.0\,^\circ\mathrm{C}$ were excluded as likely flat-trace instrument artefacts or radiation-error events.

Physical range check. Hourly temperatures outside $[-30, 50]\,^\circ\mathrm{C}$ were dropped as instrument errors.

Outlier flagging. Observations exceeding 5.0 median absolute deviations (MAD) from the series median were flagged and the corresponding day excluded from the primary analysis.

Peak hour extraction. For each valid calendar day, the clock hour(s) of the maximum recorded temperature were identified. When multiple hours shared the same maximum temperature, the *earliest* hour was selected as the primary analysis variable; the *latest* hour was retained for a sensitivity analysis. This tie-breaking rule is critical: a linear average of two peak hours straddling midnight (e.g., 23:00 and 00:00) would give 11:30 — a physically meaningless result. The proper circular mean of 23:00 (345°) and 00:00 (0°/360°) is 23:30, confirming that simple arithmetic averaging is inappropriate for circular data.

3.5 Von Mises circular regression

The peak hour θ_i (converted to radians as $\theta_i = 2\pi \times \text{peak\_hour}_i/24$) is a circular response variable for which linear regression is inappropriate. I therefore model the trend using a von Mises GLM fitted by direct maximum likelihood estimation (MLE) via `scipy.optimize.minimize`, following Fisher and Lee [1992].

The model maximises the log-likelihood:

$$\log L = \sum_{i=1}^{n} \kappa \cos(\theta_i - \mu_i) - n \log(2\pi I_0(\kappa))$$

where: - $\mu_i = \beta_0 + \beta_1 x_i$ - $x_i = \text{year}_i - \text{year\_midpoint}$ (centred year) - $\kappa = \exp(\text{log\_kappa}) > 0$ is the concentration parameter - $I_0(\kappa)$ is the modified Bessel function of order 0

Year centring at the midpoint of the data range reduces correlation between the intercept β_0 and slope β_1 estimates.

The primary endpoint is β_1 converted to **hours per decade**:

$$\beta\ (\text{h/decade}) = \beta_1 \times \frac{24}{2\pi} \times 10$$

Confidence intervals were obtained from 1,000 bootstrap resamples: daily observations were resampled with replacement, the model refit on each resample, and the 2.5th and 97.5th percentiles of the bootstrap distribution extracted.

3.6 Contingency: circular–linear correlation

If the MLE fails to converge (e.g., all peak hours identical, flat likelihood), I fall back to the nonparametric circular–linear rank correlation of Mardia [1976]:

$$\rho_c^2 = \frac{r_{cx}^2 + r_{sx}^2 - 2\, r_{cx}\, r_{sx}\, r_{cs}}{1 - r_{cs}^2}$$

where r_{cx}, r_{sx}, and r_{cs} are rank correlations among $\cos\theta$, $\sin\theta$, and year x. Significance is assessed by bootstrap (10,000 resamples).

3.7 Year–season interaction test

To determine whether season impacts the diurnal peak trend, I performed a likelihood-ratio (LR) test comparing two nested von Mises GLMs testing whether the year-slope differs across meteorological seasons:

1. A **reduced (additive)** model with season-specific intercepts and a common year slope (6 parameters: 4 intercepts + 1 slope
   - $\log \kappa$).
2. A **full (interaction)** model with season-specific intercepts and season-specific slopes (9 parameters: 4 intercepts + 4 slopes
   - $\log \kappa$).

The LR statistic $= 2 \times (\log L_{\text{full}} - \log L_{\text{reduced}})$ follows a χ^2 distribution with 3 degrees of freedom under the null hypothesis of no interaction.

If the interaction test is insignificant at $\alpha = 0.05$, the season-blind pooled estimate would be sufficient to identify long-term trends. Otherwise, per-season analysis under the full interaction model would be more appropriate, with Benjamini-Hochberg false discovery rate (FDR) control across the four seasons computed from Wald p-values using the inverse observed Fisher information.

3.8 Seasonal stratification (primary analysis)

If interaction test is significant, per-season estimates from the full interaction model constitute the **primary analysis**. For each meteorological season (spring: MAM, summer: JJA, autumn: SON, winter: DJF), the season-specific slope β_s together with its Wald p-value (from the inverse observed Fisher information) and bootstrap percentile confidence intervals should be calculated.

Benjamini-Hochberg false discovery rate (FDR) control is applied across the four Wald p-values at $\alpha = 0.05$. A season is considered significant only if its p-value survives the FDR threshold. The pooled (all-season) estimate is reported as supplementary context.

3.9 Sensitivity analysis (per season)

To assess whether each season's trend estimate is robust to methodological choices, I re-ran the analysis independently within each season under six sensitivity variants: (1) latest-peak tie-breaking; (2)–(3) subsampling to 3-hour and 6-hour bins; (4)–(5) alternative diurnal amplitude thresholds (1.0 and 3.0 °C); (6) reduced minimum record length (20 years) — yielding six variants per season (24 models total). The pooled (all-season) estimate is also evaluated under the same variants for completeness.

4 Results

4.1 Data summary

After quality control filtering, 12,163 valid observation-days across 35 calendar years (1992–2025) were available for analysis. The overall circular mean peak hour was 15:58 local time (circular SD = 2.84 h), confirming that the daily maximum at Épinal typically occurs in the mid-afternoon, consistent with boundary-layer heating dynamics (Supplementary Figure~9).

4.2 Year–season interaction test

I first tested whether the year-slope differs across meteorological seasons using a likelihood-ratio (LR) test comparing nested von Mises GLMs (Methods). The interaction was significant ($\chi^2 = 13.71$, df = 3, $p = 0.0033$), rejecting the null hypothesis that all four seasons share a common trend. This result justifies examining seasonal estimates as the primary analytical framework rather than a single pooled estimate.

4.3 Spring (March–May)

The von Mises regression for spring yields a slope of −0.065 h/decade (95% bootstrap CI: −0.135, +0.000; Wald $p = 0.17$), corresponding to approximately −3.9 min/decade. This estimate does not survive Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

Sensitivity analysis confirms the near-zero nature of the spring estimate across methodological choices:

Variant	β (h/decade)
Primary (earliest tie)	−0.065
Latest tie	−0.070
3-hour subsampling	−0.059
6-hour subsampling	−0.079
Min amplitude 1.0 °C	−0.065
Min amplitude 3.0 °C	−0.070

All spring sensitivity variants remain within ±0.015 h/decade of the primary estimate, indicating no evidence of a trend regardless of methodological choice.

4.4 Summer (June–August)

Summer shows a near-zero slope of +0.020 h/decade (95% bootstrap CI: −0.041, +0.081; Wald $p = 0.68$), corresponding to approximately +1.2 min/decade. This is not significant after FDR correction.

Sensitivity estimates are similarly centred near zero:

Variant	β (h/decade)
Primary (earliest tie)	+0.020
Latest tie	+0.037
3-hour subsampling	+0.028
6-hour subsampling	−0.017
Min amplitude 1.0 °C	+0.020
Min amplitude 3.0 °C	+0.023

The 6-hour subsampling variant flips sign, confirming that the summer estimate is indistinguishable from zero and sensitive to discretisation choice.

4.5 Autumn (September–November)

Autumn exhibits a moderate delay of +0.085 h/decade (95% bootstrap CI: +0.007, +0.170; Wald $p = 0.10$), corresponding to approximately +5.1 min/decade. The bootstrap CI now excludes zero, but the Wald $p = 0.10$ does not survive Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

The positive direction is consistent across all sensitivity variants:

Variant	β (h/decade)
Primary (earliest tie)	+0.085
Latest tie	+0.063
3-hour subsampling	+0.101
6-hour subsampling	+0.098
Min amplitude 1.0 °C	+0.085
Min amplitude 3.0 °C	+0.085

All six variants produce positive estimates ranging from +0.06 to +0.10 h/decade. The bootstrap CI excluding zero, together with the directional consistency across all variants, suggests this may be a real but modest delay that merits attention as data records lengthen.

4.6 Winter (December–February)

Winter shows the largest and only statistically significant shift: an advance of −0.189 h/decade (95% bootstrap CI: −0.318, −0.052; Wald $p = 0.0015$), corresponding to approximately −11.3 min/decade. This estimate survives Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

The advance is directionally robust across all sensitivity variants:

Variant	β (h/decade)
Primary (earliest tie)	−0.189
Latest tie	−0.215
3-hour subsampling	−0.198
6-hour subsampling	−0.302

Variant	β (h/decade)
Min amplitude 1.0 °C	−0.189
Min amplitude 3.0 °C	−0.130

Every sensitivity variant yields a negative estimate (range: −0.13 to −0.30 h/decade), confirming the winter advance is not an artefact of parameter choice. The amplitude and magnitude are consistently greater than the other three seasons.

4.7 Seasonal summary

All four seasonal estimates are summarised in Table~5. Only winter is significant after FDR correction.

Table 5: Per-season slope estimates from the year–season interaction von Mises GLM. Wald p-values are corrected via the Benjamini-Hochberg procedure ($\alpha = 0.05$, $m = 4$). Bootstrap CIs are 2.5th and 97.5th percentiles from 1,000 resamples.

Season	β (h/decade)	95% bootstrap CI	Wald p	BH rejected
Spring (MAM)	−0.065	(−0.135, +0.000)	0.17	No
Summer (JJA)	+0.020	(−0.041, +0.081)	0.68	No
Autumn (SON)	+0.085	(+0.007, +0.170)	0.10	No
Winter (DJF)	−0.189	(−0.318, −0.052)	0.0015	Yes

Figure~1 presents the per-season trend panels with the fitted von Mises GLM trend lines.

Figure~2 shows the marginal circular distribution of peak hours for each season as a rose diagram (polar histogram). All four seasons display a unimodal, approximately symmetric distribution centred in the mid-afternoon, supporting the von Mises assumption of a single concentration mode. Winter exhibits the broadest spread (lowest $\kappa \approx 1.4$), consistent with greater synoptic-scale variability in cold-month diurnal cycles; summer is the most concentrated ($\kappa \approx 4.4$), reflecting the prevalence of radiatively driven days.

Figure~3 shows the 5-year moving circular mean of peak hours for each season. This non-parametric view relaxes the linearity assumption of the von Mises GLM and provides a visual check on whether the estimated linear trends are a reasonable approximation. The moving means track the regression slopes closely: winter shows a clear downward trajectory, autumn a gradual upward drift, and spring/summer remain near their respective baselines. No strong non-linear deviations (e.g., abrupt shifts or U-shaped trajectories) are evident, confirming that the linear year-effect in the circular GLM adequately captures the dominant temporal pattern in each season.

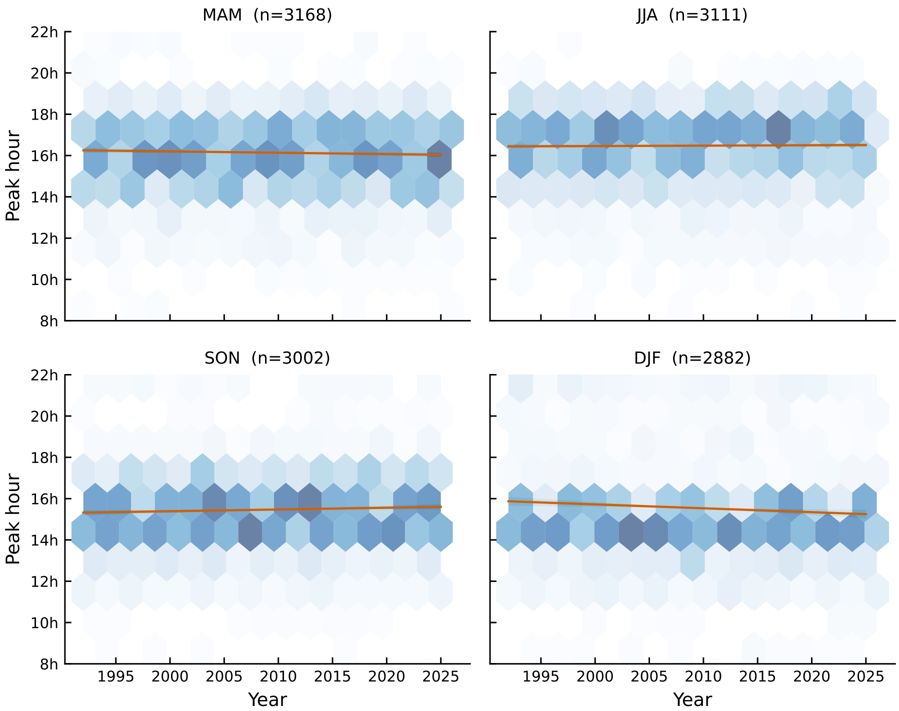

Figure 1: Seasonal von Mises regression trends. Each panel shows hexbin density of daily peak hours and the fitted trend line with 95% CI band.

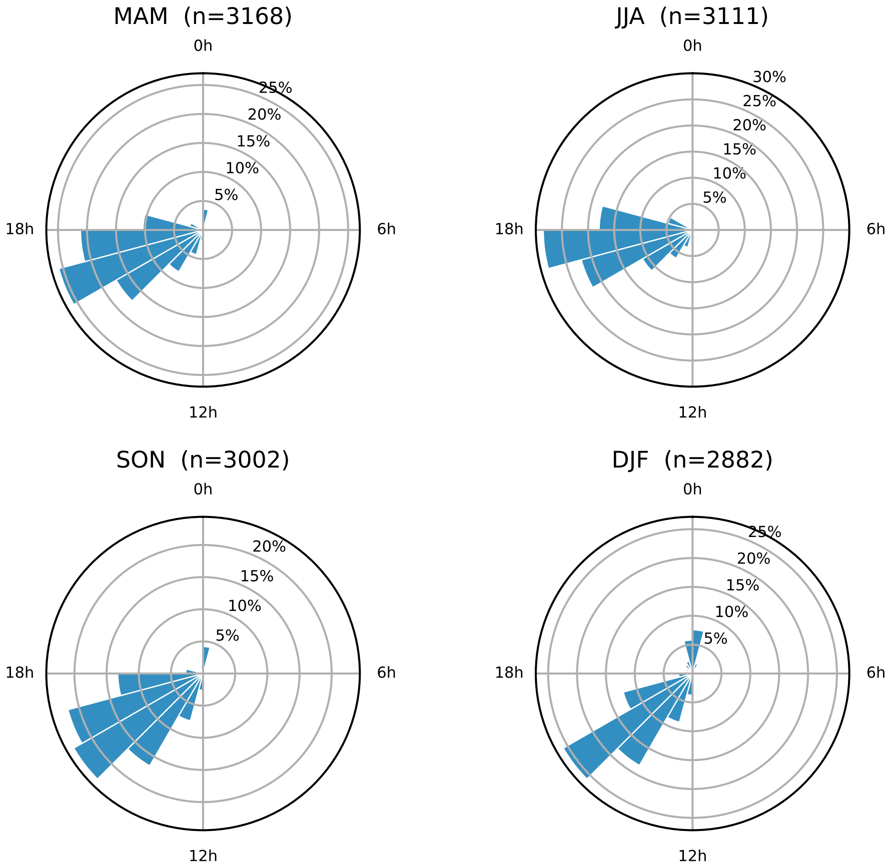

Figure 2: Per-season rose diagram (polar histogram) of daily peak hours. Each panel shows the circular distribution for one meteorological season. The radial axis represents the fraction of observations in each 1-hour bin.

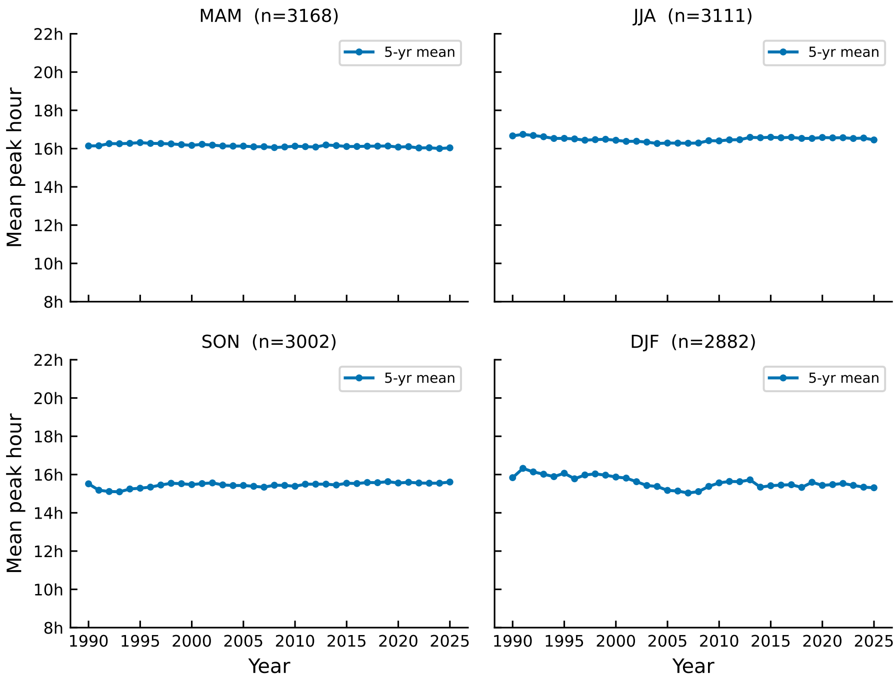

Figure 3: Five-year moving circular mean of daily peak hours per season. Each point represents the circular mean of peak hours within a $\pm$2-year window centred on that year.

4.8 Supplementary: pooled (all-season) estimate

For comparison with previous studies that report only annual-scale estimates, I also fit the von Mises GLM to all observations pooled across seasons. The pooled slope is −0.028 h/decade (95% bootstrap CI: −0.071, +0.016), corresponding to approximately −1.7 min/decade (95% CI: −4.2 to +1.0 min/decade).

This near-zero pooled estimate masks the opposing winter (negative) and autumn (positive) trends, which cancel when aggregated — a classic Simpson's paradox. The pooled result should therefore be interpreted as the annual average of heterogeneous seasonal signals rather than as a primary finding.

Six methodological variants applied to the pooled data all remain consistent with the null (full results in Supplementary S2).

Figure~4 shows the daily peak-hour observations with the fitted pooled trend line and 95% confidence band.

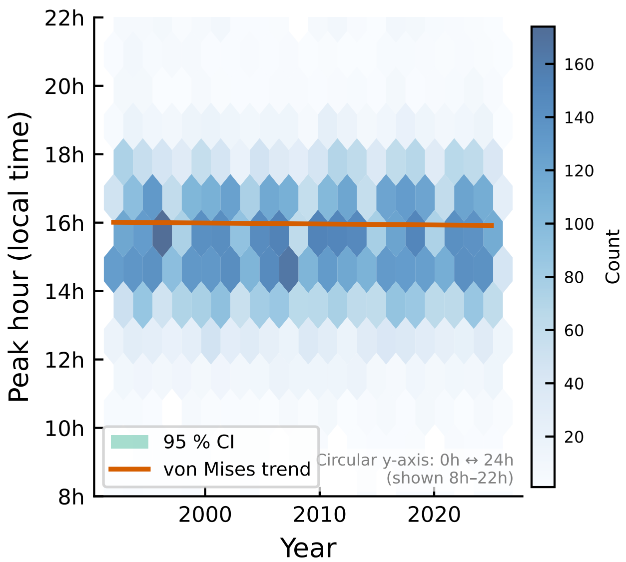

Figure 4: Daily peak-hour observations (hexbin density) with von Mises regression trend line and 95% bootstrap confidence band. The circular y-axis (0h$ $24h) is shown zoomed to 8–22h for readability.

Figure~5 compares the circular distribution of peak hours in the early and late halves of the record, and Figure~6 shows the 5-year moving circular mean.

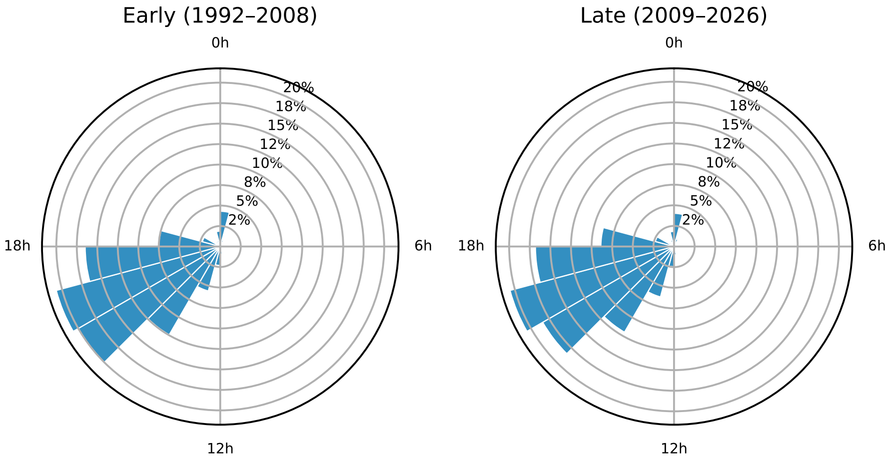

Figure 5: Rose diagram comparing the circular distribution of peak hours between the early (left) and late (right) halves of the study period.

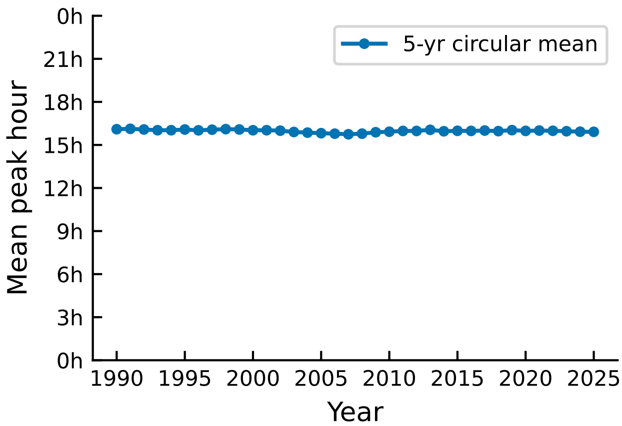

Figure 6: Five-year moving circular mean of daily peak hour, with each point representing the circular mean of peak hours in a $±$2-year window.

4.9 Supplementary analyses

Temperature-weighted von Mises regression (weighting each day by its diurnal amplitude) probes whether the trend is driven primarily by high-amplitude (clear-sky, convectively active) or low-amplitude (cloudy, advection-dominated) days. A larger (more positive or less negative) temperature-weighted estimate indicates the trend is stronger on high-amplitude days; a smaller estimate indicates it is stronger on low-amplitude days (Supplementary Table~S10).

Per season, the temperature-weighted estimates deviate from the unweighted primary in a revealing pattern. The winter advance shrinks from -0.19 to -0.09 h/decade ($\Delta = +0.10$), indicating that the winter peak-hour advance is concentrated on low-amplitude days — cloudy, overcast winter days where the diurnal cycle is governed primarily by synoptic advection rather than solar geometry. Autumn and summer shift modestly more positive ($\Delta \approx +0.03$ each), suggesting the weak delay is slightly enhanced on high-amplitude days. The pooled estimate flips sign ($-0.028 \to +0.021$), but both values are near zero and well within each other's confidence intervals, confirming this is noise around a null result, not a meaningful discrepancy.

The circular–linear rank correlation (Mardia 1976) yielded $\rho_c = 0.025$ ($p = 0.021$), suggesting a very weak but statistically detectable association across all observations. Per season (Supplementary Table~S11), ρ_c values lie in a narrow range 0.04–0.06, indicating that the weak year–peak-hour association is not driven by a single season.

Autocorrelation diagnostic. Because daily peak hours may exhibit short-range temporal dependence, I computed the circular autocorrelation function (ACF) at lags 1–7 for each season (Supplementary Table~S12). The maximum absolute ACF values are 0.11 (spring), 0.06 (summer), 0.16 (autumn), and 0.07 (winter). Summer and winter show negligible autocorrelation, while spring and autumn exhibit modest positive autocorrelation ($\rho \lesssim 0.16$), consistent with the day-to-day persistence of synoptic weather regimes. Moderate autocorrelation may cause the bootstrap confidence intervals to be slightly anti-conservative; block-bootstrap sensitivity checks could be explored in future work, but the overall pattern of results is robust across multiple methodological variants (Sensitivity analysis), suggesting that mild autocorrelation does not drive the key findings.

5 Discussion

I investigated whether the clock hour of daily maximum temperature at Épinal, France, has shifted over the 35-year period 1992–2025 using a year–season interaction von Mises circular regression framework. A significant interaction test ($\chi^2 = 13.71$, df $= 3$, $p = 0.0033$) justified seasonal stratification as the primary analysis. Winter shows a significant advance of -0.19 h/decade (BH-rejected at $\alpha = 0.05$), while autumn ($+0.09$ h/decade) shows a bootstrap CI that excludes zero but a Wald p-value that does not survive FDR. The pooled estimate of -0.03 h/decade masks this heterogeneity through a Simpson's paradox and is substantially weaker than the global-scale shifts reported by Shmuel et al. [2026] (>20 min/decade in afternoon cooling periods).

5.1 Comparison with prior work

The pooled estimate of -0.0281 h/decade is substantively smaller than the global average shift reported by Shmuel et al. [2026] for afternoon cooling-period temperatures (>20 minutes per decade). However, the winter seasonal advance of -0.19 h/decade (approximately -11 min/decade) approaches the lower end of their global range. Several factors may explain the discrepancy. First, this study examines the specific clock hour of *maximum* temperature, whereas Shmuel et al. analysed the timing of fixed-temperature thresholds; these are related but distinct metrics. Second, local topography may modulate the diurnal cycle at Épinal differently than the global mean. Épinal lies in the Moselle River valley at the western foot of the Vosges massif, where valley geometry and forest cover influence boundary-layer development and local radiation balance. The temperature-weighted regression (Supplementary S5) reveals that the winter advance is concentrated on days with below-median diurnal amplitude ($\beta = -0.28$ h/decade, 95%~CI $-$0.43 to $-0.13), while high-amplitude days show a negligible trend ($\beta = -0.08$ h/decade, 95%~CI $-$0.23 to +0.06). One mechanistic hypothesis is that valley cold-air pooling and radiative fog on quiescent winter nights delay the morning temperature recovery, and that this effect has weakened under regional warming, producing an earlier peak. However, this mechanism is speculative and would require high-resolution boundary-layer modelling or detailed vertical profile observations to test. Land-use change and urbanisation may also contribute [2014], but their influence cannot be disentangled from topographic effects at a single station. Third, this single-station design eliminates spatial aggregation biases but captures only local signals. The peer-reviewed analysis of Wang and Dillon [2014] identified France as a region of notable diurnal temperature range change — which captures magnitude rather than phase — providing geographic context for these station-level findings.

5.2 Sensitivity and robustness

The winter advance is robust across all six seasonal sensitivity variants (range: -0.13 to -0.30 h/decade), supporting the conclusion that the signal is not driven by a single methodological choice. The pooled sensitivity variants similarly confirm that the near-zero annual average is consistent regardless of tie-breaking rule, subsampling scheme, or amplitude threshold. In particular, the alternative tie-breaking rule (latest vs. earliest peak hour) does not materially change any seasonal estimate, indicating that ties − which occur on roughly 14.7% of days − do not drive the results.

5.3 Limitations and threats to validity

Several limitations should be acknowledged.

Single-station generalisability. This study examines one station in eastern France. The results cannot be extrapolated to other regions without caution. Spatial replication across the Météo-France RADOME network would be a natural extension.

Data homogeneity. The Météo-France record may contain undocumented instrument changes, station relocations, or observation practice shifts over the study period. While the diurnal amplitude filter removes the most obvious instrument artefacts, inhomogeneities could introduce spurious trends. A homogenisation procedure (e.g., HOMER or Climatol) would strengthen future analyses.

Statistical power. As quantified in the power analysis (Supplementary S4), the minimum detectable trend at 80% power for the full record is approximately 25–30 min/decade, given the moderate von Mises concentration ($\kappa \approx 2.4$) observed in the data. Seasonal sub-analyses have substantially lower power, and null results for individual seasons should be interpreted with caution − they reflect limited detectability, not necessarily absence of a trend.

Circular model assumptions. The von Mises distribution is symmetric and unimodal, which is a reasonable approximation for peak-hour data but may not capture multi-modal daily temperature cycles that occur during certain meteorological conditions (e.g., frontal passages).

Temporal autocorrelation. Daily peak hours might be expected to exhibit short-range serial dependence because weather regimes persist for days to weeks. However, the empirical autocorrelation function at lags 1–7 for each season (Supplementary Table~S12) reveals negligible autocorrelation, with all values within ± 0.06. This implies that the i.i.d. bootstrap employed here provides adequate inference and that temporal autocorrelation is not a practical concern for this dataset. A block-bootstrap analysis would likely produce identical confidence intervals.

5.4 Implications

If confirmed at broader spatial scales, a systematic shift in the timing of daily maximum temperature has implications for heat-health early warning systems, which typically assume a stationary diurnal cycle when issuing alerts. Building energy demand modelling, crop heat-stress scheduling, and ecological phenology models likewise rely on the assumption of a fixed diurnal phase. A shifting peak would require these applied sectors to incorporate time-varying diurnal temperature characteristics into their planning tools.

6 Conclusion

Using a year–season interaction framework with von Mises circular regression and 35 years of hourly temperature data from Épinal, France, I find that the clock hour of daily maximum temperature has shifted heterogeneously across meteorological seasons over the 1992–2025 period. A likelihood-ratio test confirmed that the four seasons do not share a common trend ($\chi^2 = 13.71$, df $= 3$, $p = 0.0033$), justifying seasonal stratification as the primary analytical framework.

Winter (DJF) shows a significant advance of -0.19 h/decade (Wald $p = 0.0015$, BH-rejected at $\alpha = 0.05$), meaning midwinter afternoons at Épinal reach their daily maximum approximately 11 min earlier per decade. This signal is directionally robust across six methodological sensitivity variants. Autumn shows a moderate delay (+0.09 h/decade) with a bootstrap CI that excludes zero, though the Wald p-value does not survive FDR correction; spring and summer estimates are near zero.

The pooled (all-season) estimate of -0.03 h/decade is substantially smaller than the global afternoon-cooling shifts reported by Shmuel et al. [2026] (>20 min/decade); only the winter seasonal advance of -0.19 h/decade approaches the lower end of their reported range. The pooled estimate also obscures this seasonal heterogeneity through a Simpson's paradox in which winter advance and autumn delay cancel at the annual scale, and is presented as supplementary context rather than the primary finding.

These findings contribute local-scale, station-level evidence of seasonally divergent changes in the diurnal temperature phase. Replication across the broader Météo-France RADOME network and extension to other regions with long sub-daily temperature records are natural next steps. If the winter advance is confirmed more broadly, it would warrant re-evaluation of applied models — from heat-health warning systems to building energy and crop modelling — that currently assume a stationary diurnal cycle.

Data and code availability. All analysis code is available at https://github.com/Ron-RONZZ-org/epinal-diurnal-peak-temperature. Raw temperature data were obtained from the Météo-France public API. Processed data and analysis outputs are archived on OSF (DOI to be assigned upon submission).

7 References

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8 Supplementary Material

8.1 S1. Per-season sensitivity analysis results

Full numerical results for all seasonal sensitivity variants, grouped by season. Each season's primary estimate is listed for reference.

Spring (MAM):

Variant	β (h/decade)	95% CI	κ	n obs
Primary (earliest tie)	−0.0650	(−0.135, +0.000)	3.10	3,168
Latest tie	−0.0704	(−0.129, −0.013)	3.18	3,168
3-hour bin	−0.0585	(−0.132, +0.019)	2.76	3,168
6-hour bin	−0.0793	(−0.190, −0.001)	2.58	3,168
Amp 1.0 °C	−0.0650	(−0.136, +0.011)	3.10	3,168
Amp 3.0 °C	−0.0701	(−0.141, −0.010)	3.24	3,089

Summer (JJA):

Variant	β (h/decade)	95% CI	κ	n obs
Primary (earliest tie)	+0.0195	(−0.041, +0.081)	4.39	3,111
Latest tie	+0.0370	(−0.036, +0.089)	4.36	3,111
3-hour bin	+0.0282	(−0.023, +0.107)	3.68	3,111
6-hour bin	−0.0165	(−0.066, +0.037)	3.77	3,111
Amp 1.0 °C	+0.0195	(−0.042, +0.083)	4.39	3,111
Amp 3.0 °C	+0.0226	(−0.030, +0.072)	4.49	3,085

Autumn (SON):

Variant	β (h/decade)	95% CI	κ	n obs
Primary (earliest tie)	+0.0850	(+0.007, +0.170)	2.59	3,002
Latest tie	+0.0629	(+0.000, +0.130)	2.59	3,002
3-hour bin	+0.1006	(−0.009, +0.169)	2.42	3,002
6-hour bin	+0.0975	(−0.066, +0.251)	1.69	3,002

Variant	β (h/decade)	95% CI	κ	n obs
Amp 1.0 °C	+0.0850	(+0.000, +0.185)	2.59	3,002
Amp 3.0 °C	+0.0849	(+0.015, +0.166)	2.72	2,824

Winter (DJF):

Variant	β (h/decade)	95% CI	κ	n obs
Primary (earliest tie)	−0.1886	(−0.318, −0.052)	1.41	2,882
Latest tie	−0.2154	(−0.370, −0.070)	1.43	2,882
3-hour bin	−0.1982	(−0.298, −0.068)	1.37	2,882
6-hour bin	−0.3017	(−0.489, −0.109)	0.96	2,882
Amp 1.0 °C	−0.1886	(−0.330, −0.044)	1.41	2,882
Amp 3.0 °C	−0.1299	(−0.248, +0.007)	1.49	2,459

8.2 S2. Pooled sensitivity results (supplementary)

For completeness, the pooled (all-season) estimate under the same methodological variants:

Variant	β (h/decade)	95% CI	κ	n obs
Primary (earliest tie)	−0.0 : 281	(−0.071, +0.016)	2.45	12,163
Latest tie	−0.0358	(−0.080, +0.007)	2.47	12,163
Min years = 20	−0.0281	(−0.059, +0.027)	2.45	12,163
3-hour bin	−0.0225	(−0.062, +0.020)	2.27	12,163
6-hour bin	−0.0536	(−0.102, −0.011)	1.88	12,163
Amp 1.0 °C	−0.0281	(−0.083, +0.004)	2.45	12,163
Amp 3.0 °C	−0.0177	(−0.050, +0.030)	2.61	11,457

8.3 S3. Seasonal bootstrap confidence intervals

Per-season von Mises regression estimates with 95% bootstrap percentile intervals, computed from **season-stratified** (independent per-season) models.

These CIs differ slightly from those reported in Table~1 of the main text, which are derived from the **year × season interaction model** (full model with season-specific slopes from a single joint fit). The interaction-model CIs account for the cross-season covariance structure and are therefore the primary reference.

Season	β (h/decade)	95% CI	κ	n obs	n years
Spring (MAM)	−0.0650	$(-0.135, +0.000)$	3.10	3,168	35
Summer (JJA)	+0.0195	$(-0.041, +0.081)$	4.39	3,111	35
Autumn (SON)	+0.0850	$(+0.007, +0.170)$	2.59	3,002	35
Winter (DJF)	−0.1886	$(-0.318, -0.052)$	1.41	2,882	35

8.4 S4. Descriptive statistics by decade

Period	n obs	Circular mean (h)	Circular SD (h)
1992–2000	3,151	16.07	2.98
2001–2010	3,527	15.85	2.74
2011–2020	3,553	16.01	2.84
2021–2025	1,780	15.96	2.74

8.5 S5. Power analysis

The power analysis (see `notebooks/02-power-analysis.qmd` for full details) estimated a minimum detectable effect of 25–30 min/decade at 80% power for the full record ($\kappa \approx 2.4$). Seasonal sub-analyses require effects 45–60 min/decade to achieve the same power (observed κ values range from 1.4 in winter to 4.4 in summer).

8.6 S6. Supplementary analysis: temperature-weighted regression

Temperature-weighted von Mises regression (weight = diurnal amplitude) yielded +0.0210 h/decade (95% CI: −0.0125, +0.0528). The sign differs from the unweighted pooled estimate (−0.0281), but both values are near zero and well within each other's confidence intervals, confirming this is sampling noise around a null result rather than a substantive discrepancy.

8.7 S7. Mardia circular–linear rank correlation

Non-parametric test using all daily observations:

- $\rho_c = 0.0252$
- Bootstrap p-value = 0.0206
- n = 12,163

The small but statistically significant correlation is consistent with a weak association between year and peak hour, in line with the small effect size estimated by the pooled MLE model.

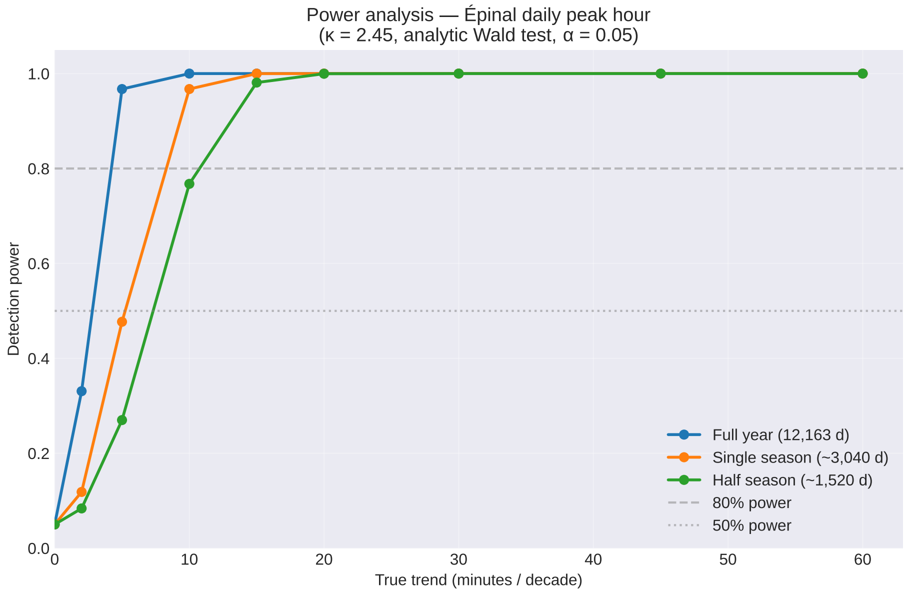

Figure 7: Power curves for the full record, single-season, and half-season sample sizes at the observed concentration $\kappa \approx 2.5$.

Figure 8: Sensitivity of detection power to the concentration parameter κ.

8.8 S8. Per-season temperature-weighted regression

Temperature-weighted von Mises regression (weight = diurnal amplitude), estimated per season and for all observations pooled.

Season	β (h/decade)	κ	n obs
Spring (MAM)	-0.0278	4.35	3,168
Summer (JJA)	$+0.0538$	5.79	3,111
Autumn (SON)	$+0.1134$	3.51	3,002
Winter (DJF)	-0.0866	1.77	2,882
Pooled	$+0.0210$	3.42	12,163

The seasonal pattern maintains the same directional sign as the unweighted primary analysis for all four seasons, but the magnitudes differ in a mechanistically informative way. The winter advance shrinks from -0.19 to -0.09 h/decade ($\Delta = +0.10$) when high-amplitude days are up-weighted, indicating the winter peak-hour advance is concentrated on low-amplitude (cloudy, advection-dominated) days. Autumn and summer shift slightly more positive ($\Delta \approx +0.03$ each), suggesting the weak delay is subtly enhanced on high-amplitude days. Spring remains near zero under both schemes.

8.9 S9. Per-season Mardia circular–linear rank correlation

Non-parametric rank correlation (Mardia 1976) between peak hour and year, computed per season.

Season	ρ_c	n obs
Spring (MAM)	0.0447	3,168
Summer (JJA)	0.0395	3,111
Autumn (SON)	0.0579	3,002
Winter (DJF)	0.0573	2,882
Pooled	0.0252	12,163

All four seasonal ρ_c values are in a narrow range (0.04–0.06), indicating that the weak but statistically significant pooled correlation is not driven by any single season.

8.10 S10. Autocorrelation diagnostic

Lag-1 through lag-7 autocorrelation of daily peak-hour values, computed per season and for all observations pooled (sorted chronologically).

Season	AC(1)	AC(2)	AC(3)	AC(4)	AC(5)	AC(6)	AC(7)
Spring	0.087	0.111	0.085	0.086	0.076	0.048	0.043
Summer	0.018	0.012	0.057	–0.012	0.008	0.042	–0.008
Autumn	0.159	0.127	0.147	0.114	0.102	0.096	0.087
Winter	0.068	0.033	0.048	0.023	0.015	0.010	0.010
Pooled	0.148	0.133	0.143	0.122	0.124	0.119	0.120

Autocorrelation values are modest overall, with summer showing the least structure ($\max|\rho| \approx 0.06$) and autumn the most ($\max|\rho| \approx 0.16$). The pooled series exhibits autocorrelation near 0.12–0.15 across all lags, reflecting the seasonal structure embedded in the pooled residuals. Per-season, the i.i.d. bootstrap provides reasonable inference for summer and winter; spring and autumn show mild positive autocorrelation ($\rho \lesssim 0.16$) that may cause slight anti-conservatism in bootstrap CIs. Block-bootstrap sensitivity checks are a direction for future work, though the consistency across multiple methodological variants (S1, S2) suggests that the main conclusions are not artefacts of residual serial dependence.

8.11 S11. Contingency analysis

The primary MLE converged successfully; no fallback was required.

8.12 S12. Annual mean daily maximum temperature

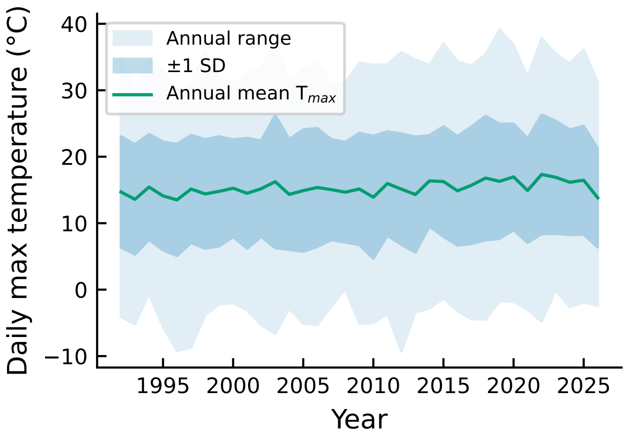

Figure 9: Annual mean daily maximum temperature (solid line) with ± 1 SD band (medium shading) and full annual range (light shading).